Circuits and Devices
Assignment no 5

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Table of Contents

Abstract 3

Objectives 3

Procedure 3

Table of Components 3

Circuit Design 4

Simulation Results 6

Without Capacitor 6

With Capacitor filter 7

Current through diode 9

Conclusion: 10

References 11

Power Supply Design on Proteus

Abstract

Objectives

- To design an unregulated power supply which inputs 220Vrms AC and outputs 12V pulsating DC at load with minimal peak-to-peak ripple voltage of 1% or less.
- Simulate the design in Proteus and verify that your designed power supply is working correctly.
- Show the current waveform from the diode, load, and input source.
- Use an appropriate regulator for the design and verify the regulation phenomena.

This project report presents the circuit design, component analysis and simulation graphs of a regulated power supply that converts a 220V AC input into a clean 12V DC output at load resistor of 120 ohm. The design includes a step-down transformer, a full-wave bridge rectifier, a smoothing/filtering capacitor, and a linear voltage regulator (LM7812). The design is simulated in Proteus to verify the proper working of the power supply, ensuring that the output voltage meets specifications with minimal ripple. Along with the voltage waveforms simulation, current through diodes is also visualized using an analog meter.

Procedure

In the process of designing this DC power supply, we used these components with the following specifications

Table of Components

Component	Model no.	Value	Ratings (Required)	Cost (In pkr)
Transformer	XFMR (ideal)	Np/Ns=18.34	2 VA	2k – 3k
AC Voltage source	VSINE	Amplitude= 311V Frequency= 50 Hz	500V	500
Diode	1N4148	1.2V (forward Vdrop)	300mA	10
Capacitor Filter	Capacitor primitive	8333µF (calculated through 1% peak to peak ripple)	40-50 V	200
Load Resistor	Analog resistor primitive	120 ohms	5-6 VA	10
Linear Regulator	LM7812	12V,1A	15V, 1A	50

Note. The references section mentions the sources/web links of these costs mentioned.

Other than these components, meters such as AC and DC voltmeters and an Oscilloscope and analog analysis graph are also used to measure and visualize the voltages/currents at different nodes of the circuit.

Circuit Design

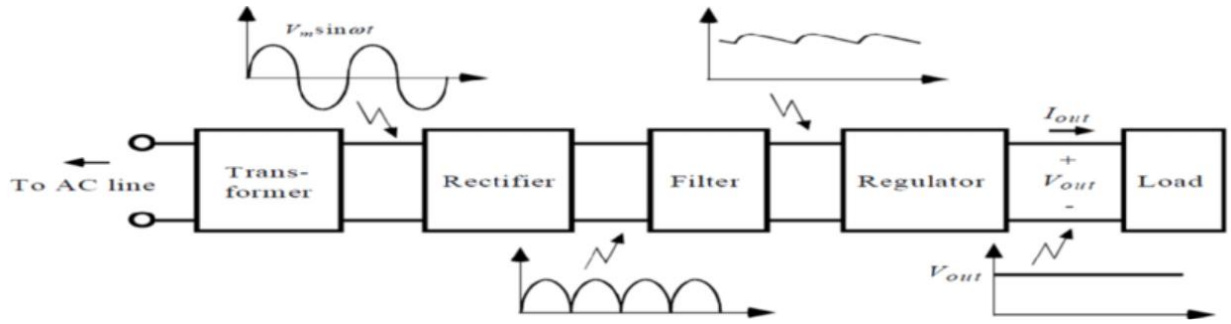


Figure 1. Block diagram of a fully regulated power supply circuit

1. An AC source of 220Vrms and 50 Hz is used. The peak voltage of source is set as 220V * (2) ^1/2 = 311V.
2. A step-down transformer is used with a turns ratio Ns/Np= 220V/12V=18.33. So, the transformer is stepping down from 220Vrms to 12Vrms as shown by the AC voltmeters in Figure 2.
3. A Full wave bridge rectifier is made using four 1N4148 diodes which will rectify input AC into DC. These diodes have a forward voltage drop of 1.2V. So, the 12V * (2) ^1/2= 16.9V input peak is subtracted by 2.4V and we get 14.57V at the output peak as shown in DC voltmeters after rectifier in Figure 2.
4. Then a filtering capacitor of 8333µF is attached in parallel to the load to reduce the ripple voltage to 1%. This specific capacitance is obtained using this formulation

$$C = \frac{I_{\text{load}}}{f \cdot V_{\text{ripple}}}$$

$I_{\text{load}} = \frac{V_{\text{dc}}}{R_{\text{Load}}} = \frac{12}{120} = 0.1\,\text{A}$

$V_{\text{ripple}} = 0.01 \times 12 = 0.12\,\text{V}$

For a full-wave rectifier, $f = 2 \times 50 = 100\,\text{Hz}$.

So, we get C = 0.1/ (100*0.12) = 8333µF.

However, a capacitor of 10,000µF can be used instead of 8333µF as it is easily commercially available. But this design is just theoretical so 8333µF is used.

The function of this smoothing capacitor is to filter out the unwanted high frequency components from the output.

5. Finally, for regulation, LM7812 is used which linearly regulates the input over-voltages at 12V DC. However, this regulator can have a voltage drop of 2V. So, the voltage to be regulated must be 14V or greater than 14V otherwise the regulation will be at less than 12V.

6. Finally, regulated voltage of 12V appears at the load resistor which is 120ohms as shown in Figure 2.

For Task 1 of the assignment, an unregulated power supply is designed as shown in Figure 2.

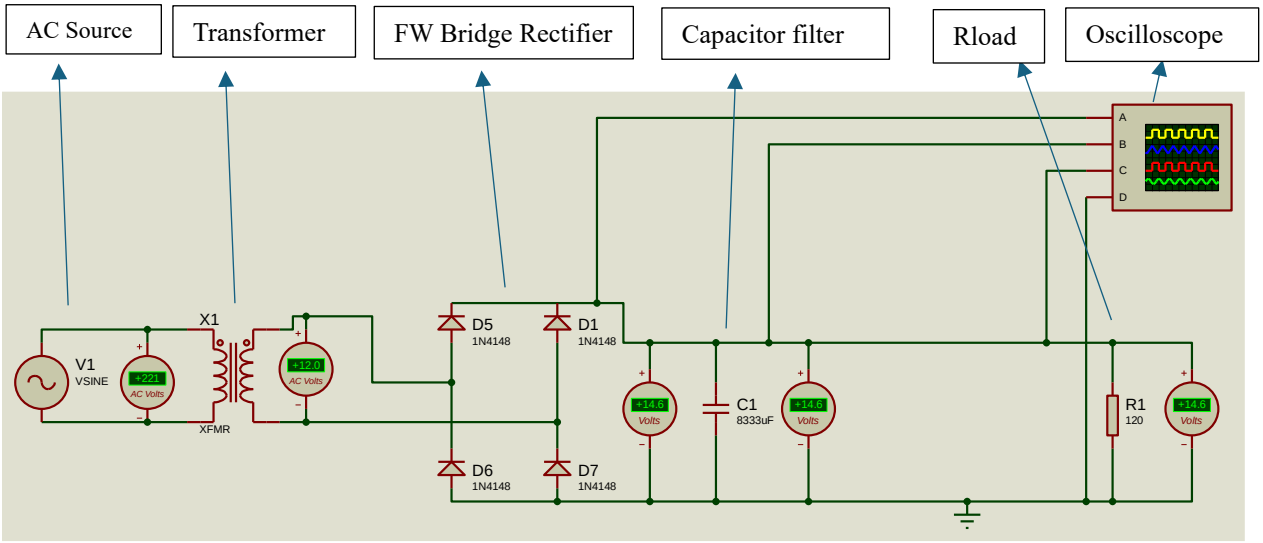


Figure 2. Circuit diagram of an unregulated power supply; Output Voltage ≠ 12V DC

For Task 2 of the assignment, a regulated power supply is designed as shown in the schematic diagram of Figure 3. All the components are same as in task 1, just a linear regulator 7812 is added between the filter and load which regulates the filtered voltage at exact 12V DC.

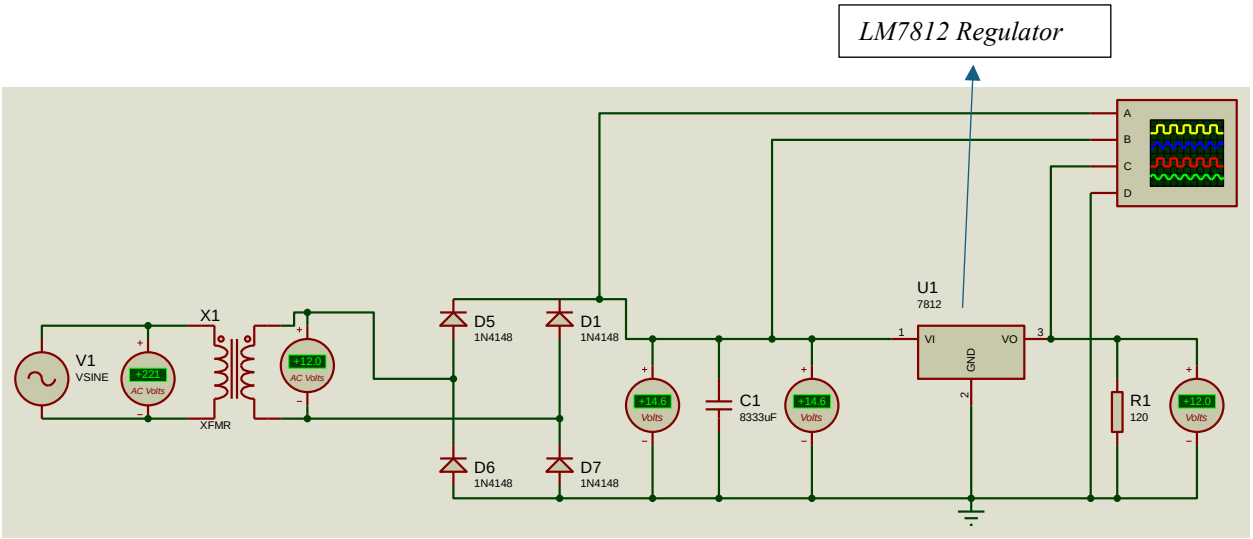


Figure 3. Circuit diagram of a regulated power supply; Output Voltage = 12V DC (fixed)

To verify the regulation phenomena, we can change the input AC voltage to any value like 250 or 300Vrms and the output DC will still be regulated at 12V. For example, the input rms voltage is

set at 280V and we got 19V filtered DC voltage which was regulated at 12V by 7812 regulator as shown in Figure 4.

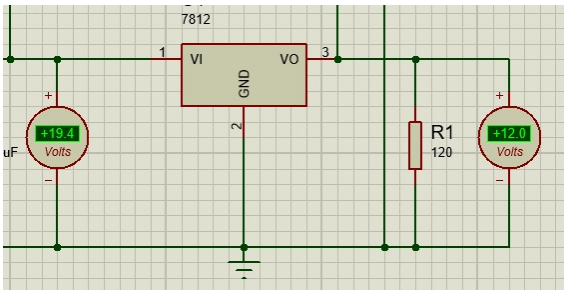


Figure 4. Voltage Regulation by 7812 regulator (19.4 to 12V)

Simulation Results

Without Capacitor

Firstly, the circuit as shown in Figure 5 is analyzed without capacitor, because with capacitor the oscilloscope's whole node becomes common and we just get the filtered voltage on all channels of oscilloscope, or all curves obtained are linear. To visualize the input to rectifier and output from rectifier separately, we are analyzing this circuit without capacitor filter. Here we are just seeing the voltage drop of 2.4V in peak input.

Note. The output to load isn't zero as shown by voltmeters here, it will be the rectified DC output such as 14.6V shown in oscilloscope results in Figure 6.

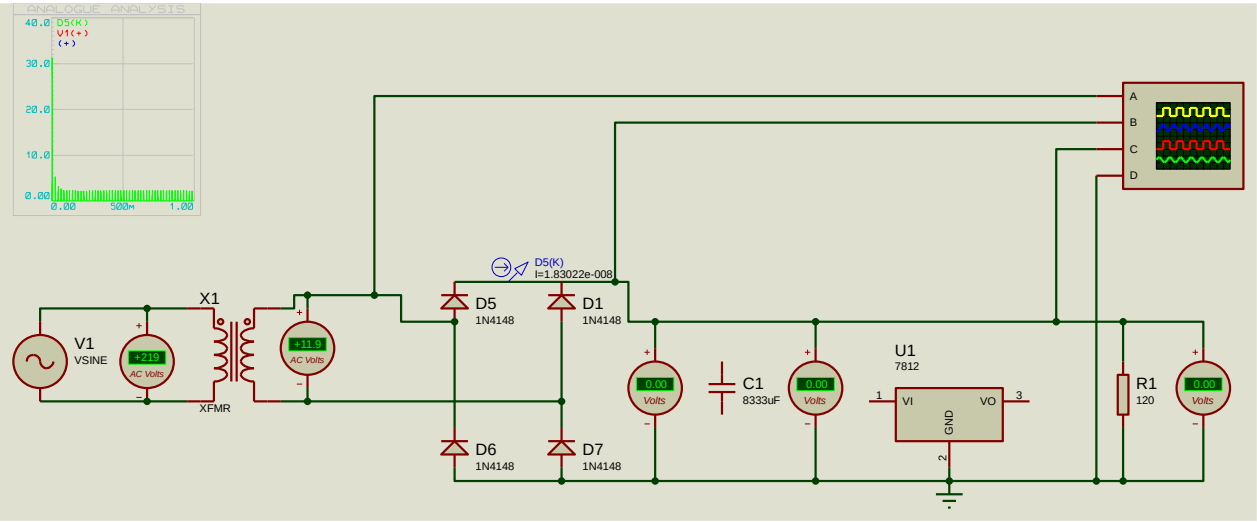


Figure 5. Circuit without capacitor/regulator

Oscilloscope output

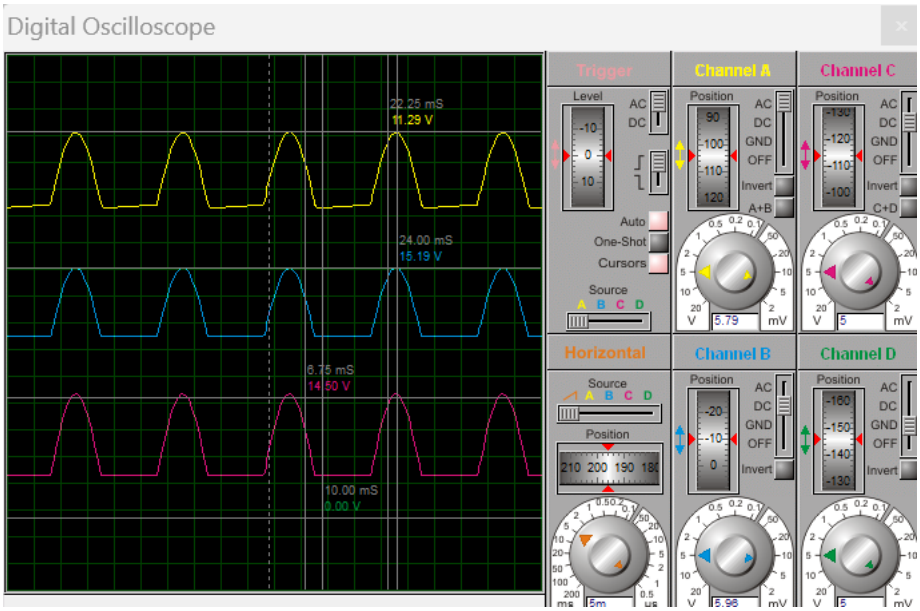


Figure 6. Results without capacitor + unregulated (Graph of Voltage vs Time period)

With Capacitor filter

Unregulated

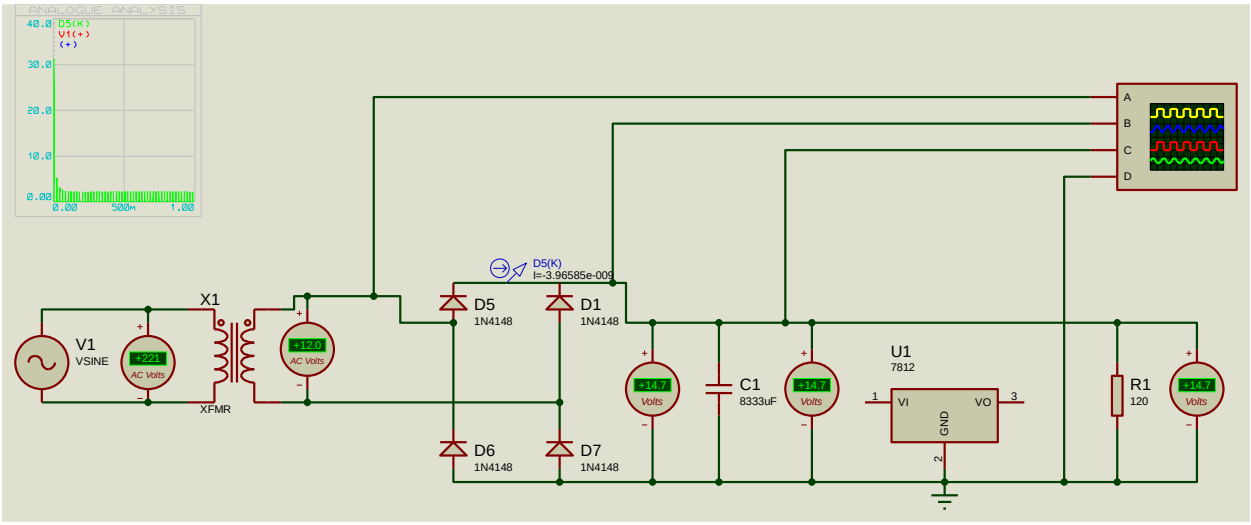


Figure 7. Unregulated power supply with filtering capacitor

We get unregulated voltage of 14.60V DC at the load with a peak-to-peak voltage ripple of just 1% as shown in simulation of Figure 8. Here the blue curve is of channel B which shows the rectified voltage while red curve shows filtered and unregulated at the load. Both of these curves have a ripple of just 0.01 so the capacitor is fulfilling our ripple requirements and providing nearly clean DC.

Oscilloscope Output

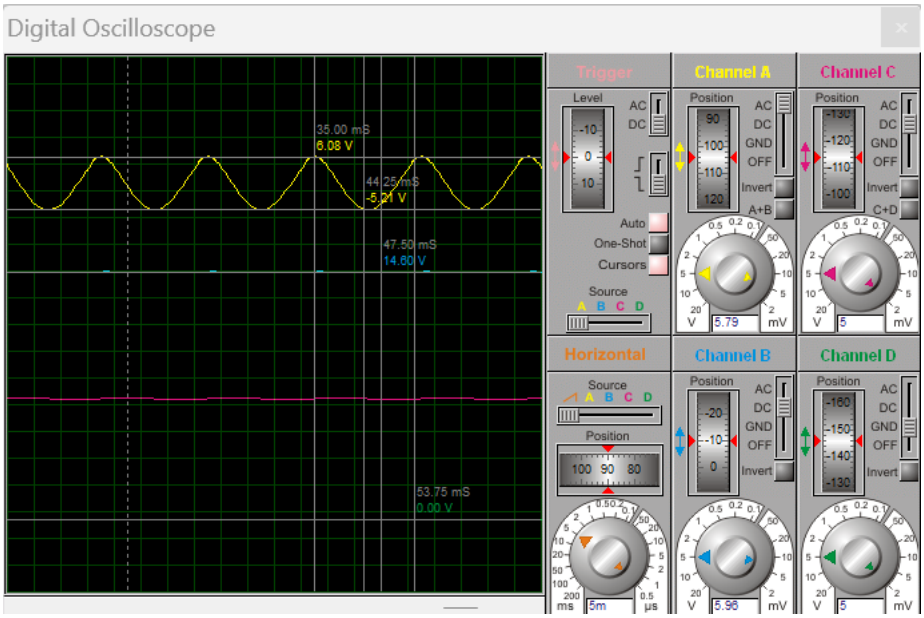


Figure 8. 'V' vs 't' graph (With capacitor/Unregulated)

Regulated

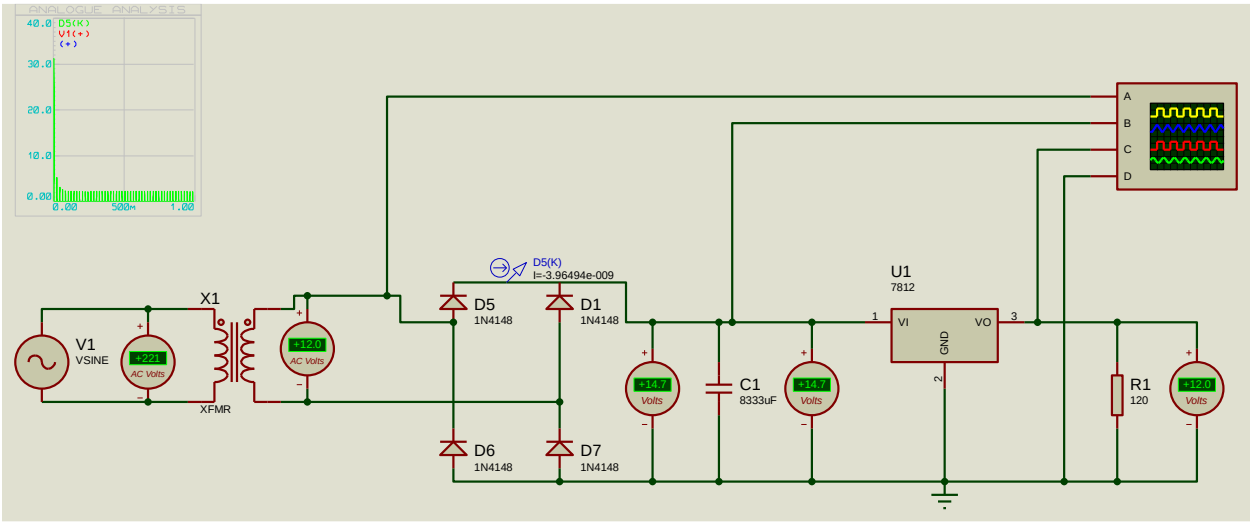


Figure 9. Regulated power supply with filtering capacitor

Now, this circuit is complete and shows 12V DC at output which is regulated i.e. if we change the input, the output voltage will remain fixed at 12V. The simulation results of this circuit can be seen from Figure 10 where Blue curve is the filtered Voltage having ripple of 0.01V while red curve is the regulated or constant DC voltage of 12V.

Oscilloscope Output

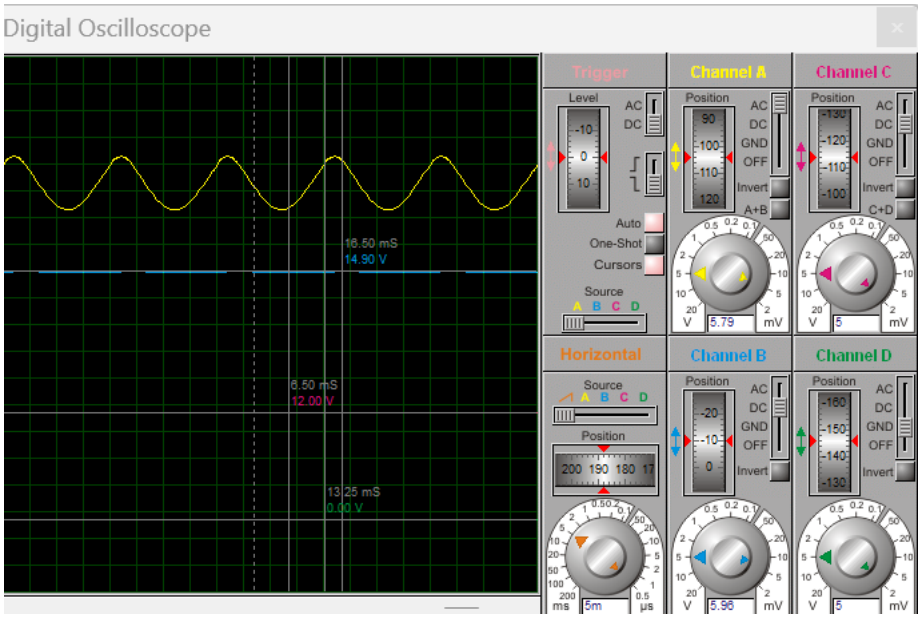


Figure 10. Results of fully regulated power supply (V vs t)

Current through diode

Current through diode is measured by placing a current probe after a diode D5 and connecting that probe to an analog analysis graph as shown in Figure 10. An Analog Graph of 'Current' vs 'Conduction time of diode' is obtained as shown in Figure 11. This Graph represents the small conduction time of diodes and the current peaks during that time. The first and highest current peak is surge current which is 30Amps at 0milli seconds. While the second peak is at 5Amps and the smaller regular peaks follow representing the short conduction of current through diode in the positive half of AC cycle only.

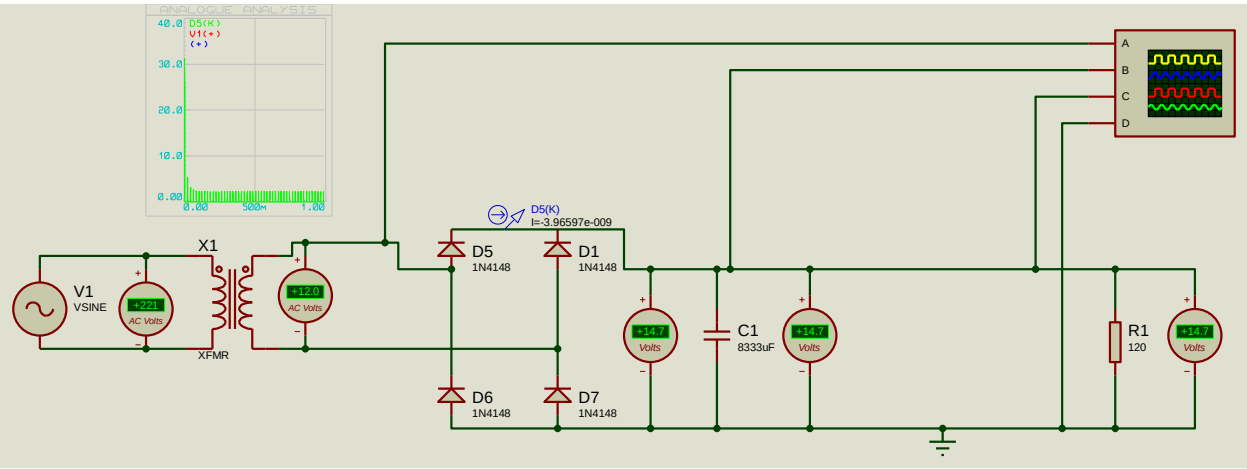


Figure 10. Representing the current probe at D5 output terminal connected to the analog graph

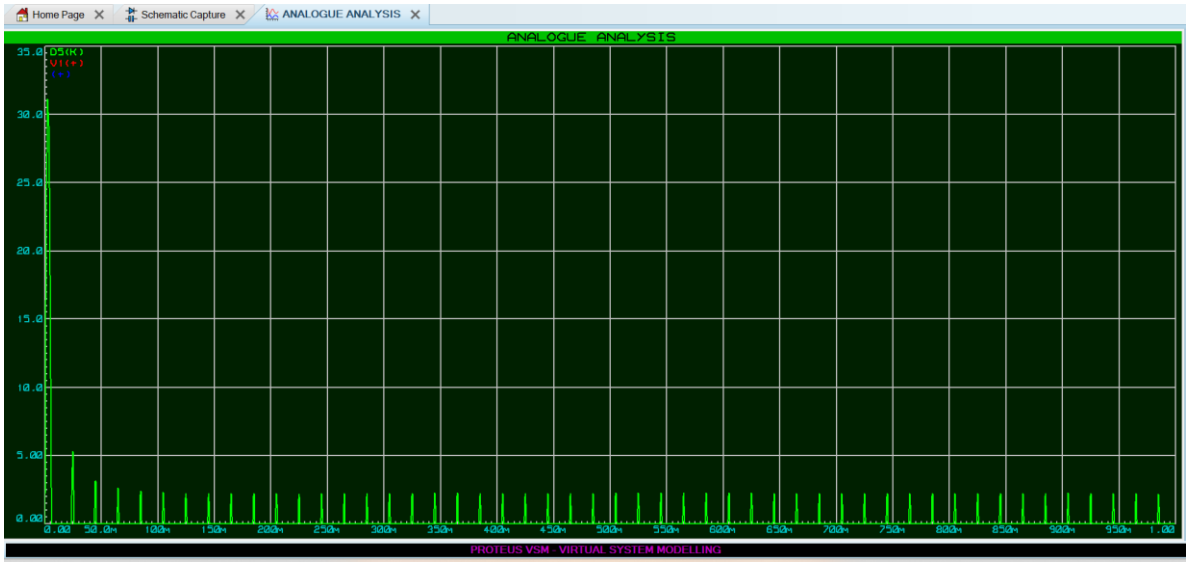


Figure 11. 'Current through diode' vs 'Conduction time of diode' graph

Moreover, the current through input and output can be measured using similar method by attaching probes there. And we can also measure current through by an indirect method of connecting a shunt resistor of probably 0.1ohm in series with any of the diodes and checking the voltage drop across resistor by connecting a voltmeter in parallel with resistor. Then by using Ohm's law, we can calculate current through diode by I=V/R. We can visualize the voltage across resistor using oscilloscope, that voltage waveform will also be the current one because voltage and current are in phase for a resistor.

Conclusion:

All the objectives of this project are successfully met. The power supply design is working properly converting 220Vrms AC into 12V clean DC. In this process, each component plays its major role. Transformer is stepping down the voltage and providing electrical isolation between the input and output which is important. The bridge rectifier is converting AC into DC, doubling the output frequency from 50 to 100Hz and posing a voltage drop of 2.4V in the input peak. While the smoothing capacitor of 8333µF is reducing the peak-to-peak voltage ripple to 1% only as asked. Finally, the 7812 regulator is maintaining the constant DC voltage of 12V at the load which is ultimate requirement. Meters used after every component are ensuring the proper functioning of our power supply. The simulation results also verify the valid working of each stage of our circuit. This project demonstrates how practical power supplies are designed for different types of loads/applications using transformers, passive filters and basic electronics devices such as rectifiers and regulators. The Power supply designed in this project was too basic, but the concepts used can be extended to advanced power supplies also.

References

Transformer cost Capacitor cost Diodes cost Voltage regulator cost